

Cultural Dimensions and Mental Health Technology: A Systematic Review of Hofstede's Dimensions in Shaping Mental Health Experiences

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ABSTRACT

This paper explores the influence of cultural factors on mental health help-seeking behaviors and the subsequent implications for the design of mental health technologies. Using Hofstede's Cultural Dimensions as a framework, we conducted a comprehensive literature review to examine how cultural variations affect patient behaviors in seeking mental health support. This review categorically analyses literature corresponding to each of Hofstede's five dimensions – Power Distance, Individualism vs. Collectivism, Masculinity vs. Femininity, Uncertainty Avoidance, and Long-Term Orientation. The findings reveal significant cultural influences on help-seeking behaviors, highlighting the need for culturally sensitive approaches in mental health technology design. This study underscores the importance of cultural awareness in the design and deployment of mental health technologies, offering insights for future research and development in this field.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

Mental Health, E-Mental Health, Culture, Cross-Cultural, Hofstede

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1 INTRODUCTION

Serious mental illnesses, including severe depression, mood disorders, and schizophrenia, are among the leading causes of long-term disability worldwide [66]. However, access to appropriate mental health care is often hindered by barriers such as social stigma, financial constraints, and long waiting periods [20, 22, 84]. Concerns about intercultural care and communication, privacy, and confidentiality further exacerbate the challenges faced by individuals seeking help. Fortunately, recent research and technological advancements have improved understanding and diagnosis of mental health illnesses, leading to more effective and accessible care that caters to patients' needs [7, 19, 84]. These technologies have demonstrated success in enhancing mental health care accessibility and providing treatment options for individuals seeking help.

Research highlights the significant impact of culture on mental health, symptom expression, and access to community support, emphasizing the need for culturally informed healthcare practices [1, 9, 35, 70, 82]. For instance, mental distress in many Asian countries is often expressed through somatic symptoms, such as stomach pain and muscle tension, contrasting with the psychological or emotional expressions common in the West [53]. Additionally, studies have found that migrants, ethnic minorities, and religious individuals tend to exhibit passive behavior during medical consultations, asking fewer questions and taking less initiative [63, 78]. These findings underscore the importance of acknowledging cultural variation in healthcare to provide effective care [62, 79, 88]. While therapists can tailor their approaches to individual needs, technology often employs a 'one-size-fits-all' approach. This oversight of diversity and cultural subtleties can lead to inaccurate diagnoses, unsuitable interventions, and a lack of personalized care [52].

Our work aims to contribute to the awareness and development of culturally appropriate mental health technology. To achieve this goal, we draw upon Hofstede's *Cultural Dimension Theory*, an established framework to describe the culture of a country based on its cultural values. [47]. For the purposes of this study, we adopt Hofstede's definition of culture, focusing on the national level to explore the impact of cultural dimensions on mental health help-seeking behaviors and technology design. Hofstede identifies five essential cultural dimensions: Individualism vs. Collectivism, Power Distance,

Masculinity vs. Femininity, Uncertainty Avoidance, and Time Orientation. These dimensions describe variations in societal values and norms, behavior, communication, and interactions within different cultural frameworks [47]. Although originally grounded in industrial psychology, Hofstede's theory is also significant in the fields of medical psychology and healthcare [11, 25, 38, 63] and has been used previously in the Human-Computer Interaction (HCI) community [33, 61, 72, 73, 81, 83, 94].

Drawing from Hofstede's theory [47], we have conducted an extensive literature review spanning medical- and design-related scientific fields. The objective of this review is to gain a comprehensive understanding of the cultural dimensions that influence mental health experiences and treatment preferences. By examining literature corresponding to each of Hofstede's dimensions, we uncover cultural influences on help-seeking behaviors and highlight the need for culturally sensitive approaches in mental health technology design.

2 RELATED WORK

Recent advancements within the HCI community have highlighted the critical impact of gender and cultural factors on mental illness manifestations, prompting a shift towards culturally sensitive approaches in technology design [5, 23, 92]. Ayobi et al.'s [5] method of using visual cards to meet diverse needs, alongside Williams' [92] investigation into care disparities, emphasizes the growing need for culturally informed digital solutions. Further enriching this dialogue, Yellowlees et al. [95] propose a framework that echoes Hofstede's dimensions, offering strategies to enhance cultural adaptiveness in technologies.

Despite considerable progress, the integration of cultural insights into the design of technologies, particularly for mental health applications, is still in its infancy. Most research has been centered around interface design, leveraging Hofstede's cultural dimensions to assess and adapt cross-cultural design differences across various websites [13, 21, 43, 59, 60, 64, 68, 72, 73]. These studies have primarily examined specific UI attributes—such as imagery, color schemes, and navigation styles (linear vs. nonlinear). Notably, the work of Reinecke & Bernstein [72] presents a comprehensive method for dynamically adjusting interfaces to cultural expectations, demonstrating the practical utility of Hofstede's dimensions in technology design. While these contributions mark significant steps towards UI adaptability, their focus has been limited to interface characteristics, validated mostly through small-scale academic testing [73], thus highlighting a research gap in the broader context of mental health technology design [92, 96]. Our study aims to bridge this gap by building upon the foundation laid by Reinecke & Bernstein [72, 73], exploring how Hofstede's dimensions influence mental health-seeking behaviors and inform broader design considerations for technology.

While existing research has begun to explore cultural impacts on technology behaviors beyond UI design, such efforts often focus on specific instances, like the adoption of contact tracing apps [36], contact tracing preferences [57], dynamics within digital friendship networks [81], and patterns in app store feedback [27]. These studies demonstrate how particular cultural dimensions, most notably

Individualism, Power Distance and Uncertainty Avoidance, influence digital engagement and acceptance of technology. However, the scope of the aforementioned studies is generally confined to one or two dimensions, revealing a gap in the literature for research that encompasses all of Hofstede's dimensions in a holistic manner, especially within the realm of mental health technology.

3 METHODOLOGY

To synthesize prior research on the impact of culture on behaviors related to mental health help-seeking, we conducted a systematic literature review. Selecting appropriate search terms presented a challenge – overly general terms yielded an excessive number of papers, while highly specific terms risked missing relevant literature. We created our corpus using keyword searches on article indexes including IEEE Xplore, ACM Digital Library, and Scopus:

- **Interface Design Keywords:** ALL[culture] AND ALL[interface] AND ALL[Hofstede].
- **Behavioral Design Keywords:** TITLE-ABS-KEY[Any of: Culture dimensions, Individualism-collectivism, Masculinity-femininity, Power distance, Time Orientation, Uncertainty avoidance] AND ALL[Any of: Medical communication, Mental Health, Medicine].

We manually inspected the first 100 search results obtained for behavioral keywords and the first 50 results for interface considerations. This process entailed analyzing the article titles and abstracts to assess their relevance to our research objectives. These cut-off points were established based on team capacity, the diminishing relevance of articles further down the search results, and prior experience in literature review processes, which suggested that the majority of pertinent articles are often found within the initial set of results. This pragmatic approach allowed us to efficiently identify articles of significant relevance while ensuring the review remained focused within the constraints of our resources. Papers not aligning with our review's focus were excluded at this stage. Subsequently, we conducted a backward citation search, examining the reference lists of the remaining relevant papers to uncover additional pertinent literature. This approach resulted in a total of 72 sources that were pertinent for our comprehensive review. During the screening process, two authors independently evaluated each paper's relevance, resolving disagreements through discussion and consensus. Our data analysis involved a thematic coding process specific to each of Hofstede's cultural dimensions. We analyzed each paper for every dimension to identify themes related to mental health help-seeking behaviors and technology design.

In this paper, we present the main themes drawn from our analysis at a level which we believe is most appropriate for communicating the influence of cultural differences on health-seeking behaviors and articulating their potential implications for the design of mental health support technologies. The full results of the analysis are too extensive to include in this paper but can be found in Appendix A organized by cultural dimension, fully detailing the themes and implications under each. The subsequent sections of this paper will discuss these dimension-specific themes at a higher level to highlight their relevance for mental health technology design.

4 RESULTS

Our systematic literature review revealed distinct themes related to each of Hofstede's cultural dimensions, shedding light on their influences on mental health help-seeking behaviors and implications for design. It is necessary, however, to recognize that the cultural dimensions are not fixed or exclusively tied to a particular culture, as initially suggested in Hofstede's original work. There exists a range of characteristics among individuals from the same country, indicative of a multitude of influences and the diversity inherent within any culture. Our findings center on behavioral design implications, examining how culture influences mental health help-seeking. Although we give some consideration to interface design, our primary emphasis is on unraveling the impact of cultural factors on mental health help-seeking behaviors and culturally specific design considerations. More specific interface implications, though important, have been addressed in some detail in prior literature [21, 59, 64, 68, 72].

Individualism vs Collectivism

Hofstede's Individualism vs Collectivism (IvC) dimension [47] illustrates the societal balance between personal goals and group objectives. Individualistic cultures (e.g. USA, Australia, Great Britain [65]) prioritize self-reliance and personal achievements, focusing on individual rights and needs. In contrast, collectivistic cultures (e.g. China, Guatemala [65]) emphasize group goals, interdependence, and community loyalty. Our review identified four key themes in mental health related to IvC:

- **Orientation:** Individuals from individualistic cultures tend to favor an independent self and patient-centered care with a preference for individualized approaches. Individuals from collectivistic cultures tend to value interdependence, with family and interpersonal relationships significantly influencing health decisions [47, 87].
- **Autonomy and Help-Seeking Behaviors:** Individuals from individualistic cultures tend to prefer autonomy in health management and are more open to sharing personal information with practitioners. Individuals from collectivistic cultures tend towards social integration and are often hesitant to seek external help, influenced by family dynamics and mental health stigma [2, 10, 23, 71, 97].
- **Information Exchange:** Individuals from individualistic cultures tend to lean towards direct communication, sharing only essential information. Individuals from collectivistic cultures tend to prefer a familial approach in their interactions with healthcare providers. They prefer doctors who are supportive, caring, and incorporate social talk and emotions [77].
- **Views on Digital Support:** Individuals from individualistic cultures tend to have smaller support networks and may hesitate to seek help for personal issues, while individuals from collectivistic cultures tend to show lower compliance to mental health interventions, possibly due to social stigma and shame [4, 23, 23, 51, 55].

Our review suggests that for individualistic cultures, which tend to value independence and personal achievement, technology should support self-reliance [86]. Features like personalized

alerts and efficient, static digital platforms that offer direct information and communication can align with their preferences for autonomy and succinct communication. Conversely, collectivistic cultures tend to prioritize interdependence and community; technology designs for them could focus more on fostering social connections [97]. This includes leveraging social support networks, incorporating collaborative activities, and adding privacy features to address the potential reluctance for external help-seeking. Furthermore, collectivistic cultures tend to resonate with engaging design elements like gamification which could foster increased adherence. Interfaces that incorporate personalization and a sense of warmth can foster a feeling of care and community, resonating with the collectivist preference for healthcare interactions that are more personal and family-oriented.

Power Distance

Hofstede's Power distance (PD) dimension addresses how societies manage power inequality. High PD cultures (e.g. Malaysia, Guatemala, Panama [65]) embrace hierarchical structures with a clear authority gradient, often showing reluctance to challenge higher-status individuals. In contrast, low PD cultures (e.g. Austria, Israel, Denmark [65]) value equality, advocating for even power distribution and encouraging questioning of authority. Our review identified four key themes related to PD in the context of mental health and medical help-seeking:

- **Power Inequality:** Individuals from high PD cultures tend to exhibit strong respect for medical authority, while individuals from low PD cultures tend to advocate for a balance between professional guidance and community input. [45, 48, 91].
- **Expectations:** In high PD cultures, therapists are expected to adopt a directive approach, whereas in low PD cultures, there is a preference for collaborative decision-making [2, 29, 41, 46].
- **Self-Disclosure:** Discussing private health issues can be challenging for individuals from high PD cultures with a preference for privacy, while individuals from low PD cultures tend to be more open to transparent communication [2].
- **Collaboration:** Patients from high PD cultures often tend to view doctors as ultimate authorities, whereas in patients from low PD cultures, there tends to be a focus on joint doctor-patient exploration of health issues [77, 91].

Our review suggests that for individuals from high PD cultures who tend to highly respect medical authorities, mental health technologies should emphasize medical expertise in technological interventions, focusing on content led by healthcare professionals. These designs should respect hierarchical structures in healthcare and ensure privacy, especially accommodating family involvement. Conversely, for individuals from low PD cultures who tend to prefer a more egalitarian approach, technologies could balance professional advice with community input. This aligns with their value of open communication and equal information exchange in healthcare. Collaborative features could be adapted to each PD context. In high PD cultures, design strategies should aim to bridge the gap between patients and doctors, fostering a more collaborative

environment. Conversely, in low PD cultures, the focus should shift towards enhancing joint decision-making and mutual exploration between doctors and patients, promoting an environment of shared responsibility and insight.

Masculinity vs Femininity

The Masculinity vs Femininity (MvF) dimension contrasts cultural preferences for traits traditionally seen as masculine, such as assertiveness, with those considered feminine, such as nurturing. Masculine cultures (e.g. Japan, Austria, Italy [65]) focus on achievement and distinct gender roles, while feminine cultures (e.g. Sweden, Norway, Netherlands [65]) emphasize life quality and collaborative problem-solving¹. The MvF dimension has been less explored in research than the other dimensions partly due to criticisms of reinforcing traditional gender stereotypes and not adequately reflecting the evolving and diverse nature of gender roles across cultures [50]. However, Hofstede defends this dimension by emphasizing its basis in societal roles and values rather than individual gender identities, arguing that it captures broader cultural preferences rather than prescribing specific gender behaviors. Our review identified three key themes in mental health related to MvF:

- **Attitudes:** Masculine cultures tend to exhibit a reluctance to seek medical care, associating unhealthy behaviors with strength and self-reliance. Feminine cultures are inclined to demonstrate a more positive attitude towards help-seeking and holistic approaches [3, 18, 31, 75].
- **Approach:** In masculine cultures, there is a preference for action-focused interventions and practical problem-solving strategies. Feminine cultures are inclined to value emotional support from healthcare professionals [12, 28].
- **Engagement:** Attention and adherence in masculine culture is often driven by competition and games, while feminine cultures prioritize mutual exchange and emotional support [45, 59].

Our review suggests that for masculine-oriented cultures, mental health technologies should focus on supporting healthy behaviors and behavior change, emphasizing strength and self-reliance. Practical utility and action-oriented strategies should be central, incorporating elements like decision-support tools and case studies. Gamification could be used to enhance attention and adherence [59]. In contrast, for feminine-oriented cultures, the design should prioritize environments, either physical or digital, that foster active listening, empathy, and emotional support. Technologies could facilitate building trusting relationships in which individuals feel understood.

Uncertainty Avoidance

The Uncertainty Avoidance (UA) dimension examines how cultures manage uncertainty and ambiguity. High UA cultures (e.g. Greece, Portugal, Guatemala [65]) prefer structured environments with clear rules and predictability, while low UA cultures (e.g. Singapore, Jamaica, Denmark [65]) are more comfortable with ambiguity and risk-taking. Our review identified four key themes in mental health related to UA:

- **Trust & Innovation:** Low UA cultures tend to show openness to innovative treatment approaches, embracing new technologies and solutions with curiosity. Conversely, high UA cultures are inclined have a lower tolerance for ambiguity, preferring established protocols and traditional methods, often exhibiting lower trust in new technologies [8, 30, 47, 56, 59, 63].
- **Proactiveness:** Low UA cultures may adopt a 'wait-and-see' approach, overlooking symptoms and avoiding prescription medications. In contrast, high UA cultures tend to exhibit high anxiety and stress levels, seeking extensive information and explanations to alleviate uncertainty and minimize risks [11, 24, 25, 59, 63, 89].
- **Communication:** In low UA cultures, there is a focus on eye contact and rapport-building in doctor-patient interactions, facilitating open communication. High UA cultures may feel uncomfortable in direct conversations about health, leading to less expressive consultations [63, 89].
- **Empowerment:** Low UA cultures are inclined to attribute health outcomes to personal abilities, whereas high UA cultures may feel more powerless and attribute outcomes to external factors [47].

Our review suggests that for low UA cultures, which tend to value flexibility and curiosity, mental health technologies should emphasize innovation and user autonomy. Design features that encourage proactive health management and educate users in recognizing symptoms and making informed decisions could be beneficial. These tools can foster curiosity and support individuals in managing their health proactively. Conversely, for high UA cultures, which tend to prefer structure and predictability, technology could focus more on building trust and security. Designs could aim to demonstrate the effectiveness and reliability of health interventions, potentially incorporating user reviews and case studies to foster trust. Comprehensive health education features and tools that allow users to confidently manage their health information could be crucial.

Time Orientation

The Time Orientation (TO) dimension assesses how societies prioritize long-term commitments versus short-term gratification. Long TO cultures (e.g. China, Japan, South Korea [65]) value future rewards, perseverance, and maintaining traditions, while short TO cultures (e.g. Pakistan, Nigeria, Philippines [65]) focus on immediate gratification and individual success. Our review identified six themes:

- **Attribution:** Long TO cultures tend to attribute success to effort and failure to a lack thereof, suggesting a focus on personal responsibility in health outcomes. Conversely, short TO cultures might attribute both failure and success to external factors such as luck or fate, potentially impacting their motivation for sustained care [47, 48, 67].
- **Preventive Care:** Long TO cultures tend to be more proactive in follow-up care and self-screening, recognizing the importance of continuous self-care. Short TO cultures might

¹'Masculine' and 'Feminine' here represent cultural attitudes related to gender roles, not biological sex.

be less engaged in such practices without immediate symptoms, often overlooking the benefits of preventive measures [37, 76].

- **Habits:** Long TO cultures are inclined to adopt health promoting activities, such as regular exercise and balanced diets. Among short TO cultures there may be a tendency towards health risks for short-term pleasure, such as smoking or unhealthy eating [40, 42, 58, 76].
- **Traditions:** Among long TO cultures, traditions tend to be seen as adaptable which may translate to a greater acceptance of innovative health approaches. Short TO cultures, who are inclined to value established traditions, may prefer more structured health guidance that aligns with their longstanding beliefs [15, 47, 48].
- **Social Network:** Long TO cultures tend to value interpersonal relations and social networks as resources for health success. Short TO cultures, which may focus more on maintaining social appearance, might prefer discreet and private health interventions to avoid visibility or scrutiny [48, 59, 65].
- **Causality:** Long TO cultures often tend to focus on practical aspects of the disease ('What' and 'How'), whereas short TO cultures may be more interested in understanding the 'Why' behind symptoms, indicating a preference for in-depth exploration [48].

Our review suggests that for individuals from long TO cultures, mental health technology designs could support proactive health management. Designs should foster health-promoting habits [42], provide tools for informed decision-making, and accommodate innovative approaches to health. Flexibility and openness to innovation are key, as these cultures tend to be more receptive towards experimental and holistic methods. Conversely, for individuals from short TO cultures, designs could focus on structured guidance that aligns with established traditions. Emphasizing immediate and long-term outcomes, motivating adherence through engaging platforms, and respecting the preference for private health management is key. Designs should also cater to a deeper exploration of health issues, highlighting the importance of understanding the 'Why' behind symptoms.

5 DISCUSSION

Our findings illuminate the critical intersection between culture and mental health, in line with previous work [5, 92, 96]. They highlight the role cultural factors play in shaping both patient behavior and their importance for the development of mental health technologies. We observed that cultural nuances are pivotal in determining individuals' approaches to mental health care, their willingness to self-disclose, and their healthcare interaction preferences. This underscores the importance of a cultural lens in mental health technology design, ensuring alignment with the cultural values and needs of diverse user groups.

Our proposed design considerations for each cultural dimension are intended to spur deeper reflection and ideation, rather than serve as prescriptive solutions. They illustrate that cultural alignment in mental health technology does not necessarily mean devising entirely distinct solutions for each cultural group. Instead,

it often involves adjusting the framing, focus, and features of interventions to resonate with the values of different cultures. This approach aims to inspire practitioners and designers to engage critically with the role of culture in technology design, using these insights as a foundation for broader exploration and understanding.

To operationalize Hofstede's Cultural Dimension Framework [47], we drew inspiration from Reinecke & Bernstein's work [72], which exemplifies how frameworks can make the integration of culture into technology design more structured and tangible. Our ongoing work involves the development of design cards to facilitate the incorporation of cultural factors into the design process for mental health technologies. These cards will describe each dimension, highlighting its impact, and offer concrete design suggestions based on our findings, inspirational examples from the research community, or reflective questions to guide designers. Targeting HCI researchers and UX/UI designers in the mental health space, our proposed cards, aim to bridge the research-practice gap, as previously proven to be an effective translational tools [17, 32]. The intention is to make incorporating culture a more conscious and tangible aspect of the design phase, supporting the development of culturally sensitive technologies and general cultural awareness. After the development of the cards, we plan to refine them through workshops with our target audience, ensuring they are practical and effective in fostering culturally inclusive design solutions.

Moreover, this approach opens the door for creating adaptable and personalized technology solutions. For instance, following Reinecke & Bernstein's [72] method, interfaces and interventions could automatically adjust to align with the cultural expectations and needs of patients based on their backgrounds. This adaptability extends to allowing healthcare practitioners to configure tools and interventions more independently during treatment, tailoring them to the specific cultural context of each patient. This adaptability mirrors the 'Cycle of Care' framework in medicine, as described by Pendleton et al. [69] which underscores the importance of adapting to the evolving values, beliefs, and attitudes of patients in each consultation. Just as this framework advocates for responsiveness in clinical interactions, our research promotes a similar consideration in mental health technology design.

6 LIMITATIONS

A limitation of our approach is the reliance on Hofstede and Minkow's country-level dimensions [65], available online². It is important to recognize that cultural values can vary significantly within countries, necessitating a more individualized and intersectional approach. Future work could focus on eliciting these cultural dimensions at a more individual level.

Additionally, our analysis was based on a diverse array of existing sources that employed various methodologies. This heterogeneity in approaches could potentially lead to inconsistencies in the data from which our insights are drawn. While employing a single, unified approach in an empirical study might offer more consistency, such a method would be impractical for covering the same breadth of cultures, practices, and attitudes as our extensive literature review. Furthermore, a singular empirical approach might overlook valuable findings already existing in the literature, which

²<https://www.hofstede-insights.com/country-comparison-tool>

have been captured through diverse research methods. This trade-off highlights the complexity of researching cultural dimensions in mental health and underscores the need for a balanced approach that combines both breadth and depth of investigation.

7 CONCLUSION

Our research underscores the significance of cultural dimensions in shaping mental health help-seeking behaviors and technology design. Each of the five cultural dimensions presents unique challenges and opportunities in the context of mental health care. Our findings highlight that a one-size-fits-all approach in mental health technology design is inadequate. Instead, acknowledging and integrating these cultural nuances can lead to more effective and user-friendly technologies. This involves the application of these insights and their continuous exploration and refinement. The integration of interdisciplinary perspectives, involving psychology, anthropology, and human-computer interaction, is crucial in this endeavor. We envision mental health technologies as not only intervention tools but as bridges of cultural understanding, contributing to a more inclusive and holistic approach to mental well-being.

REFERENCES

- [1] Nazanin Andalibi, Oliver L. Haimson, Munmun De Choudhury, and Andrea Forte. 2016. Understanding Social Media Disclosures of Sexual Abuse Through the Lenses of Support Seeking and Anonymity. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 3906–3918. <https://doi.org/10.1145/2858036.2858096>
- [2] Lena Andary, Yvonne Stolk, and Steven Klimidis. 2003. *Assessing mental health across cultures*. Australian Academic Press.
- [3] Rebecca P Ang, Kam Ming Lim, Ai-Girl Tan, and Tow Yee Yau. 2004. Effects of gender and sex role orientation on help-seeking attitudes. *Current Psychology* 23 (2004), 203–214.
- [4] Julian K. Ayeh, Norman Au, and Rob Law. 2013. Predicting the intention to use consumer-generated media for travel planning. *Tourism Management* 35 (2013), 132–143. <https://doi.org/10.1016/j.tourman.2012.06.010>
- [5] Amid Ayobi, Annali Grimes, Sue Mackinnon, Ewan Soubutts, Rachel Eardley, Zoe Banks Gross, Rachael Gooberman-Hill, and Aisling Ann O'Kane. 2021. Designing Visual Cards for Digital Mental Health Research with Ethnic Minorities. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference* (Virtual Event, USA) (DIS '21). Association for Computing Machinery, New York, NY, USA, 942–954. <https://doi.org/10.1145/3461778.3462085>
- [6] Wayland Barber and Albert N. Badre. 1998. Culturability: the merging of culture and usability.
- [7] Jakob E. Bardram, Mads Frost, Károly Szántó, Maria Faurholt-Jepsen, Maj Vinberg, and Lars Vedel Kessing. 2013. Designing Mobile Health Technology for Bipolar Disorder: A Field Trial of the Monarca System. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (CHI '13). Association for Computing Machinery, New York, NY, USA, 2627–2636. <https://doi.org/10.1145/2470654.2481364>
- [8] Carol M. Barnum. 2011. 10 - International usability testing. In *Usability Testing Essentials*, Carol M. Barnum (Ed.). Morgan Kaufmann, Boston, 319–354. <https://doi.org/10.1016/B978-0-12-375092-1.00010-6>
- [9] Kamaldeep Bhui, Stephen Stansfeld, Jenny Head, Mary Haines, Sheila Hillier, Stephanie Taylor, Russell Viner, and Robert Booy. 2005. Cultural identity, acculturation, and mental health among adolescents in east London's multiethnic community. *Journal of epidemiology and community health* 59 (05 2005), 296–302. <https://doi.org/10.1136/jech.2003.014456>
- [10] Stephen Bochner. 1994. Cross-cultural differences in the self concept: A test of Hofstede's individualism/collectivism distinction. *Journal of cross-cultural psychology* 25, 2 (1994), 273–283.
- [11] Michael A Borg. 2012. National cultural dimensions as drivers of inappropriate ambulatory care consumption of antibiotics in Europe and their relevance to awareness campaigns. *The Journal of antimicrobial chemotherapy* 67 3 (2012), 763–7.
- [12] Suzanne Brownhill, Kay Wilhelm, Lesley Barclay, and Gordon Parker. 2002. Detecting depression in men: A matter of guesswork. *International Journal of Mens Health* 1 (2002), 259–280.
- [13] Inga Burgmann, Philip J. Kitchen, and Russell Williams. 2006. Does culture matter on the web. *Marketing Intelligence & Planning* 24 (2006), 62–76.
- [14] Ewa Callahan. 2005. Cultural Similarities and Differences in the Design of University Web sites. *Journal of Computer-Mediated Communication* 11, 1 (2005), 239–273. <https://doi.org/10.1111/j.1083-6101.2006.tb00312.x> arXiv:<https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1083-6101.2006.tb00312.x>
- [15] Monta Chayakonvikom, Parin Fuangvut, and Stephen Cannell. 2016. Exploring Education Culture by Employing Hofstede's Cultural Dimensions to Evaluate the Effectiveness of the Current ERP Training Approach in Thailand. *Journal of Education and Training Studies* 4, 10 (2016), 79–89.
- [16] Sejung Marina Choi, Wei-Na Lee, and Hee-Jung Kim. 2005. LESSONS FROM THE RICH AND FAMOUS: A Cross-Cultural Comparison of Celebrity Endorsement in Advertising. *Journal of Advertising* 34 (2005), 85–98.
- [17] Lucas Colusso, Cynthia L. Bennett, Gary Hsieh, and Sean A. Munson. 2017. Translational Resources: Reducing the Gap Between Academic Research and HCI Practice. In *Proceedings of the 2017 Conference on Designing Interactive Systems* (Edinburgh, United Kingdom) (DIS '17). Association for Computing Machinery, New York, NY, USA, 957–968. <https://doi.org/10.1145/3064663.3064667>
- [18] Will H Courtenay. 2000. Engendering health: A social constructionist examination of men's health beliefs and behaviors. *Psychology of men & masculinity* 1, 1 (2000), 4.
- [19] David Coyle, Nicola McGlade, Gavin Doherty, and Gary O'Reilly. 2011. Exploratory Evaluations of a Computer Game Supporting Cognitive Behavioural Therapy for Adolescents. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Vancouver, BC, Canada) (CHI '11). Association for Computing Machinery, New York, NY, USA, 2937–2946. <https://doi.org/10.1145/1978942.1979378>
- [20] Charles E. Cunningham, John R. Walker, John D. Eastwood, Henny Westra, Heather Rimas, Yvonne Chen, Madalyn Marcus, Richard P. Swinson, Keyna Bracken, and The Mobilizing Minds Research Group. 2014. Modeling Mental Health Information Preferences During the Early Adult Years: A Discrete Choice Conjoint Experiment. *Journal of Health Communication* 19, 4 (2014), 413–440. <https://doi.org/10.1080/10810730.2013.811324> arXiv:<https://doi.org/10.1080/10810730.2013.811324> PMID: 24266450
- [21] Dianne Cyr. 2013. Website Design, Trust and Culture: An Eight Country Investigation. *Electron. Commer. Rec. Appl.* 12, 6 (nov 2013), 373–385. <https://doi.org/10.1016/j.elerap.2013.03.007>
- [22] Kate Daine, Keith Hawton, Vinod Singaravelu, Anne Stewart, Sue Simkin, and Paul Montgomery. 2013. The Power of the Web: A Systematic Review of Studies of the Influence of the Internet on Self-Harm and Suicide in Young People. *PLoS ONE* 8 (2013).
- [23] Munmun De Choudhury, Sanket S. Sharma, Tomaz Logar, Wouter Eekhout, and René Clausen Nielsen. 2017. Gender and Cross-Cultural Differences in Social Media Disclosures of Mental Illness. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing* (Portland, Oregon, USA) (CSCW '17). Association for Computing Machinery, New York, NY, USA, 353–369. <https://doi.org/10.1145/2998181.2998220>
- [24] Marieke De Mooij. 2019. Consumer behavior and culture: Consequences for global marketing and advertising. *Consumer Behavior and Culture* (2019), 1–472.
- [25] Reginald Deschepper, Robert H Vander Stichele, and Flora M Haaier-Ruskamp. 2002. Cross-cultural differences in lay attitudes and utilisation of antibiotics in a Belgian and a Dutch city. *Patient Education and Counseling* 48, 2 (2002), 161–169. [https://doi.org/10.1016/S0738-3991\(02\)00017-4](https://doi.org/10.1016/S0738-3991(02)00017-4)
- [26] Claire Dormann and Cristina Chisalita. 2002. Cultural values in web site design. In *Proceedings of the 11th European Conference on Cognitive Ergonomics ECCE11*.
- [27] Michael D Dzandu, Buddhi Pathak, and Sergio de Cesare. 2022. Acceptability of the COVID-19 contact-tracing app—does culture matter? *Government Information Quarterly* 39, 4 (2022), 101750.
- [28] Carol Emslie, Damien Ridge, Sue Ziebland, and Kate Hunt. 2007. Exploring men's and women's experiences of depression and engagement with health professionals: More similarities than differences? A qualitative interview study. *BMC family practice* 8 (02 2007), 43. <https://doi.org/10.1186/1471-2296-8-43>
- [29] Sussie Eshun and Regan AR Gurung. 2009. Introduction to culture and psychopathology. *Culture and mental health: Sociocultural influences, theory, and practice* (2009), 1–17.
- [30] C. M. Nadeem Faisal, Martin Gonzalez-Rodriguez, Daniel Fernandez-Lanvin, and Javier de Andres-Suarez. 2017. Web Design Attributes in Building User Trust, Satisfaction, and Loyalty for a High Uncertainty Avoidance Culture. *IEEE Transactions on Human-Machine Systems* 47, 6 (2017), 847–859. <https://doi.org/10.1109/THMS.2016.2620901>
- [31] Hannah Farrimond. 2012. Beyond the caveman: Rethinking masculinity in relation to men's help-seeking. *Health* 16, 2 (2012), 208–225.
- [32] Anton Fedosov, Masako Kitazaki, William Odom, and Marc Langheinrich. 2019. Sharing Economy Design Cards. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3290605.3300375>

- [33] Ricarda Anna-Lena Fischer, Rita Walczuch, and Emitza Guzman. 2021. Does culture matter? impact of individualism and uncertainty avoidance on app reviews. In *Proceedings of the 43rd International Conference on Software Engineering: Software Engineering in Society* (Virtual Event, Spain) (ICSE-SEIS '21). IEEE Press, 67–76. <https://doi.org/10.1109/ICSE-SEIS52602.2021.00016>
- [34] Gabrielle Ford and Paula Kotzé. 2005. Designing Usable Interfaces with Cultural Dimensions. In *Human-Computer Interaction - INTERACT 2005*, Maria Francesca Costabile and Fabio Paternò (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 713–726.
- [35] Elsa A Friis-Healy, Gabriela A Nagy, and Scott H Kollins. 2021. It Is Time to REACT: Opportunities for Digital Mental Health Apps to Reduce Mental Health Disparities in Racially and Ethnically Minoritized Groups. *JMIR Ment Health* 8, 1 (26 Jan 2021), e25456. <https://doi.org/10.2196/25456>
- [36] Sarah Geber and Shirley S Ho. 2023. Examining the cultural dimension of contact-tracing app adoption during the COVID-19 pandemic: A cross-country study in Singapore and Switzerland. *Information, Communication & Society* 26, 11 (2023), 2229–2249.
- [37] Léontine Goldzahl. 2017. Contributions of risk preference, time orientation and perceptions to breast cancer screening regularity. *Social Science & Medicine* 185 (2017), 147–157. <https://doi.org/10.1016/j.socscimed.2017.04.037>
- [38] Narayan Gopalkrishnan. 2018. Cultural diversity and mental health: Considerations for policy and practice. *Frontiers in public health* 6 (2018), 308538.
- [39] E.W. Gould, N. Zalcaria, and S.A.M. Yusof. 2000. Applying culture to Web site design: a comparison of Malaysian and US Web sites. In *18th Annual Conference on Computer Documentation. ipcc sigdoc 2000. Technology and Teamwork. Proceedings. IEEE Professional Communication Society International Professional Communication Conference an.* 161–171. <https://doi.org/10.1109/IPCC.2000.887273>
- [40] Lori C. Guthrie, Stephen C. Butler, and Michael M. Ward. 2009. Time perspective and socioeconomic status: A link to socioeconomic disparities in health? *Social Science & Medicine* 68, 12 (2009), 2145–2151. <https://doi.org/10.1016/j.socscimed.2009.04.004>
- [41] Regina Hechanova and Lynn Waelde. 2017. The influence of culture on disaster mental health and psychosocial support interventions in Southeast Asia. *Mental Health, Religion & Culture* 20, 1 (2017), 31–44. <https://doi.org/10.1080/13674676.2017.1322048> arXiv:https://doi.org/10.1080/13674676.2017.1322048
- [42] James M Henson, Michael P Carey, Kate B Carey, and Stephen A Maisto. 2006. Associations among health behaviors and time perspective in young adults: Model testing with boot-strapping replication. *Journal of behavioral medicine* 29 (2006), 127–137.
- [43] Marc Hermeking. 2005. Culture and Internet consumption: Contributions from cross-cultural marketing and advertising research. *Journal of computer-mediated communication* 11, 1 (2005), 192–216.
- [44] Marc Hermeking. 2005. Culture and Internet Consumption: Contributions from Cross-Cultural Marketing and Advertising Research. *Journal of Computer-Mediated Communication* 11, 1 (2005), 192–216. <https://doi.org/10.1111/j.1083-6101.2006.tb00310.x> <https://academic.oup.com/jcmc/article-pdf/11/1/192/22233942/jcmc.com0192.pdf>
- [45] Geert Hofstede. 1984. *Culture's consequences: International differences in work-related values*. Vol. 5. sage.
- [46] Geert Hofstede. 2001. *Culture's consequences: Comparing values, behaviors, institutions and organizations across nations*. sage.
- [47] Geert Hofstede. 2003. Cultural dimensions. www.geert-hofstede.com (2003).
- [48] Geert Hofstede. 2011. Dimensionalizing Cultures: The Hofstede Model in Context. *Online readings in psychology and culture* 2, 1 (2011), 2307–0919.
- [49] Geert Hofstede, Gert Jan Hofstede, and Michael Minkov. 2005. *Cultures and organizations: Software of the mind*. Vol. 2. McGraw-hill New York.
- [50] Geert H Hofstede and Geert Hofstede. 1998. *Masculinity and femininity: The taboo dimension of national cultures*. Vol. 3. Sage.
- [51] Yoojung Kim, Dongyoung Sohn, and Sejung Marina Choi. 2011. Cultural difference in motivations for using social network sites: A comparative study of American and Korean college students. *Computers in Human Behavior* 27, 1 (2011), 365–372. <https://doi.org/10.1016/j.chb.2010.08.015> Current Research Topics in Cognitive Load Theory.
- [52] Arthur Kleinman. 1982. Neurasthenia and depression: A study of somatization and culture in China. *Culture, medicine and psychiatry* 6, 2 (1982), 117–190.
- [53] Arthur Kleinman, Joan Kleinman, and B Good. 2007. Somatization: The interconnections in Chinese society among culture, depressive experiences, and the meanings of pain. *Beyond the body proper: Reading the anthropology of material life* (2007), 468–74.
- [54] Anett Kralisch, Martin Eisend, and Bettina Berendt. 2005. The Impact of Culture on Website Navigation Behaviour.
- [55] Yi-Chieh Lee, Yichao Cui, Jack Jamieson, Wayne Fu, and Naomi Yamashita. 2023. Exploring Effects of Chatbot-Based Social Contact on Reducing Mental Illness Stigma. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 488, 16 pages. <https://doi.org/10.1145/3544548.3581384>
- [56] Veronica Liljander, Filippa Gillberg, Johanna Gummerus, and Allard Van Riel. 2006. Technology readiness and the evaluation and adoption of self-service technologies. *Journal of Retailing and Consumer Services* 13, 3 (2006), 177–191.
- [57] Xi Lu, Eunkyung Jo, Seora Park, Hwajung Hong, Yunan Chen, and Daniel A Epstein. 2022. Understanding Cultural Influence on Perspectives Around Contact Tracing Strategies. *Proceedings of the ACM on Human-Computer Interaction* 6, CSCW2 (2022), 1–26.
- [58] Noreen E Mahon, Thomas J Yarcheski, and Adela Yarcheski. 1997. Future time perspective and positive health practices in young adults: An extension. *Perceptual and Motor Skills* 84, 3_suppl (1997), 1299–1304.
- [59] Aaron Marcus. 2010. Cross-cultural user-interface design for work, home, play, and on the way. In *ACM SIGGRAPH ASIA 2010 Courses*. 1–160.
- [60] Aaron Marcus and Valentina-Johanna Baumgartner. 2004. A practical set of culture dimensions for global user-interface development. In *Lecture Notes in Computer Science*. Springer Berlin Heidelberg, Berlin, Heidelberg, 252–261.
- [61] Aaron Marcus and Emilie West Gould. 2000. Crosscurrents: Cultural Dimensions and Global Web User-Interface Design. *Interactions* 7, 4 (jul 2000), 32–46. <https://doi.org/10.1145/345190.345238>
- [62] Rodrigo Mariño, Mike Morgan, and Matthew Hopcraft. 2012. Transcultural dental training: addressing the oral health care needs of people from culturally diverse backgrounds. *Community dentistry and oral epidemiology* 40 Suppl 2 (10 2012), 134–40. <https://doi.org/10.1111/j.1600-0528.2012.00733.x>
- [63] Ludwien Meeuwesen, Atie van den Brink-Muinen, and Geert Hofstede. 2009. Can dimensions of national culture predict cross-national differences in medical communication? *Patient Education and Counseling* 75, 1 (2009), 58–66. <https://doi.org/10.1016/j.pec.2008.09.015>
- [64] SHAFIE BIN MEHAD. 2008. Inducting the Dimensions of Islamic Culture: A Theoretical Building Approach and Website Information Architecture Design Application. (2008).
- [65] Michael Minkov and Geert Hofstede. 2012. Hofstede's fifth dimension: New evidence from the World Values Survey. *Journal of cross-cultural psychology* 43, 1 (2012), 3–14.
- [66] Christopher J. L. Murray, Alan D Lopez, World Health Organization, World Bank, and Harvard School of Public Health. 1996. The Global burden of disease : a comprehensive assessment of mortality and disability from diseases, injuries, and risk factors in 1990 and projected to 2020 : summary / edited by Christopher J. L. Murray, Alan D. Lopez. , Published by the Harvard School of Public Health on behalf of the World Health Organization and the World Bank pages.
- [67] Jeff Niederdeppe and Andrea Gurmankin Levy. 2007. Fatalistic beliefs about cancer prevention and three prevention behaviors. *Cancer Epidemiology Biomarkers & Prevention* 16, 5 (2007), 998–1003.
- [68] Sehrish Nizamani, Khalil-u-Rehman Khoubati, Sarwat Nizamani, Shahzad Memon, Saad Nizamani, and Nazish Basir. 2018. Exploring the Impact of Cultural Preferences in Website Design: A Controlled Experiment Based on Three University Websites of Pakistan. In *2018 International Conference on Information and Communication Technology for the Muslim World (ICT4M)*. 329–334. <https://doi.org/10.1109/ICT4M.2018.00067>
- [69] David Pendleton, Theo Schofield, Peter Tate, and Peter Havelock. 2003. *The new consultation: developing doctor-patient communication*. OUP Oxford.
- [70] Sachin R. Pendse, Kate Niederhoffer, and Amit Sharma. 2019. Cross-Cultural Differences in the Use of Online Mental Health Support Forums. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW, Article 67 (nov 2019), 29 pages. <https://doi.org/10.1145/3359169>
- [71] Rhys Price-Robertson, Angela Obradovic, and Brad Morgan. 2017. Relational recovery: beyond individualism in the recovery approach. *Advances in Mental Health* 15, 2 (2017), 108–120.
- [72] Katharina Reinecke and Abraham Bernstein. 2013. Knowing What a User Likes: A Design Science Approach to Interfaces That Automatically Adapt to Culture. *MIS Q.* 37, 2 (jun 2013), 427–454. <https://doi.org/10.25300/MISQ/2013/37.2.06>
- [73] Katharina Reinecke and Abraham Bernstein. 2013. Knowing what a user likes: a design science approach to interfaces that automatically adapt to culture. *MIS Q.* 37, 2 (jun 2013), 427–454. <https://doi.org/10.25300/MISQ/2013/37.2.06>
- [74] Andrew Reyes, Rose Constantino, Chad Cross, Rhigel Tan, and Judzia Bombard. 2020. Resilience, Trauma, and Cultural Norms Regarding Disclosure of Mental Health Problems among Foreign-Born and US-Born Filipino American Women. *Behavioral Medicine* 46 (10 2020), 217–230. <https://doi.org/10.1080/08964289.2020.1725413>
- [75] Aaron B Rochlen, Debora A Paterniti, Ronald M Epstein, Paul Duberstein, Lindsay Willeford, and Richard L Kravitz. 2010. Barriers in diagnosing and treating men with depression: a focus group report. *American journal of men's health* 4, 2 (2010), 167–175.
- [76] Angelica M Roncancio, Kristy K Ward, and Maria E Fernandez. 2014. The influence of time perspective on cervical cancer screening among Latinas in the United States. *Journal of health psychology* 19, 12 (2014), 1547–1553.
- [77] Sanne Schinkel, Barbara C Schouten, Fatmagül Kerpelick, Bas Van Den Putte, and Julia CM Van Weert. 2018. Perceptions of barriers to patient participation: are they due to language, culture, or discrimination? *Health communication* (2018).

- [78] Sanne Schinkel, Barbara C Schouten, Fatmagül Kerpçelik, Bas van den Putte, and Julia C. M. van Weert. 2018. Perceptions of Barriers to Patient Participation: Are They Due to Language, Culture, or Discrimination? *Health Communication* 34 (2018), 1469 – 1481.
- [79] Barbara C Schouten, Antoon Cox, Gözde Duran, Koen Kerremans, Leyla Köseoglu Banning, Ali Lahdidioui, Maria van den Muijsenbergh, Sanne Schinkel, Hande Sungur, Jeanine Suurmond, et al. 2020. Mitigating language and cultural barriers in healthcare communication: Toward a holistic approach. *Patient Education and Counseling* 103, 12 (2020), 2604–2608.
- [80] Greg Scott, Joseph Ciarrochi, and Frank P Deane. 2004. Disadvantages of being an individualist in an individualistic culture: Idiocentrism, emotional competence, stress, and mental health. *Australian Psychologist* 39, 2 (2004), 143–154.
- [81] Agrima Seth, Jiyin Cao, Xiaolin Shi, Ron Dotsch, Yozen Liu, and Maarten W. Bos. 2023. Cultural Differences in Friendship Network Behaviors: A Snapchat Case Study. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (<conf-loc>, <city>Hamburg</city>, <country>Germany</country>, </conf-loc>) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 671, 14 pages. <https://doi.org/10.1145/3544548.3581074>
- [82] Sang-Wha Sien, Shalini Mohan, and Joanna McGrenere. 2022. Exploring Design Opportunities for Supporting Mental Wellbeing Among East Asian University Students in Canada. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 330, 16 pages. <https://doi.org/10.1145/3491102.3517710>
- [83] Denise Sogemeier, Yannick Forster, Frederik Naujoks, Josef F. Krems, and Andreas Keinath. 2022. How to Map Cultural Dimensions to Usability Criteria: Implications for the Design of an Automotive Human-Machine Interface. In *Adjunct Proceedings of the 14th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Seoul, Republic of Korea) (AutomotiveUI '22). Association for Computing Machinery, New York, NY, USA, 123–126. <https://doi.org/10.1145/3544999.3554786>
- [84] Katarzyna Stawarz, Chris Preist, Deborah Tallon, Laura Thomas, Katrina Turner, Nicola Wiles, David Kessler, Roz Shafran, and David Coyle. 2020. Integrating the Digital and the Traditional to Deliver Therapy for Depression: Lessons from a Pragmatic Study. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376510>
- [85] Constantine Stephanidis. 2009. User Interfaces for All: Concepts, Methods, and Tools.
- [86] Nancy Tomes. 2006. The Patient As A Policy Factor: A Historical Case Study Of The Consumer/Survivor Movement In Mental Health. *Health affairs (Project Hope)* 25 (05 2006), 720–9. <https://doi.org/10.1377/hlthaff.25.3.720>
- [87] Harry C Triandis. 1989. The self and social behavior in differing cultural contexts. *Psychological review* 96, 3 (1989), 506.
- [88] Kelsy Lin Ulrey and Patricia Amason. 2001. Intercultural Communication Between Patients and Health Care Providers: An Exploration of Intercultural Communication Effectiveness, Cultural Sensitivity, Stress, and Anxiety. *Health Communication* 13, 4 (2001), 449–463. https://doi.org/10.1207/S15327027HC1304_06
- [89] Anju Verma, Ann Griffin, Jane Dacre, and Andrew Elder. 2016. Exploring cultural and linguistic influences on clinical communication skills: a qualitative study of international medical graduates. *BMC Medical Education* 16 (2016), 1–10.
- [90] Thomas Vöhringer-Kuhnt. 2001. *Kulturelle Einflüsse auf die Gestaltung von Mensch-Maschine Systemen*. GRIN Verlag.
- [91] Yinyu Wang, Yanfen Gu, and Haiping Yu. 2022. Correlation between patients' power distance and their willingness to participate in patients' safety: A cross-sectional study. *Journal of Nursing Management* 30, 5 (2022), 1345–1354.
- [92] Lucretia Ann Williams. 2023. *Design and Evaluation of Culturally Responsive Digital Mental Health Technology for Racial-Ethnic Minorities*. Ph.D. Dissertation. University of California, Irvine.
- [93] Anna Xygykou, Panote Siriaraya, Alexandra Covaci, Holly Gwen Prigerson, Robert Neimeyer, Chee Siang Ang, and Wan-Jou She. 2023. The "Conversation" about Loss: Understanding How Chatbot Technology Was Used in Supporting People in Grief. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 646, 15 pages. <https://doi.org/10.1145/3544548.3581154>
- [94] Judith Yaaqoubi and Katharina Reinecke. 2018. The Use and Usefulness of Cultural Dimensions in Product Development. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (<conf-loc>, <city>Montreal QC</city>, <country>Canada</country>, </conf-loc>) (CHI EA '18). Association for Computing Machinery, New York, NY, USA, 1–9. <https://doi.org/10.1145/3170427.3174368>
- [95] Peter Yellowlees, Shayna Marks, Don Hilty, and Jay H. Shore. 2008. Using e-Health to Enable Culturally Appropriate Mental Healthcare in Rural Areas. *Telemedicine and e-Health* 14, 5 (2008), 486–492. <https://doi.org/10.1089/tmj.2007.0070> arXiv:https://doi.org/10.1089/tmj.2007.0070 PMID: 18578685.
- [96] Peter Yellowlees, Shayna Marks, Don Hilty, and Jay H Shore. 2008. Using e-health to enable culturally appropriate mental healthcare in rural areas. *Telemedicine and e-Health* 14, 5 (2008), 486–492.
- [97] Sung-Kyung Yoo. 1996. *Individualism-collectivism, attribution styles of mental illness, depression symptomatology, and attitudes toward seeking professional help: A comparative study between Koreans and Americans*. University of Minnesota.

A CULTURAL DIMENSION TABLES

Individualism vs. Collectivism		
	Individualism	Collectivism
Behavioural Design Implication		
Autonomy Orientation	Individualist-oriented individuals prioritize an independent self and patient-centered care [49, 87]. Design interventions can explore individualized approaches and self-help strategies as alternatives to community building and relationships [86].	Collectivist-oriented individuals value an interdependent self [49]. Design interventions may explore recognizing the significance of family and interpersonal relationships, potentially incorporating these elements throughout the design process [97].
	Individualists place a strong emphasis on independence and personal achievement [71]. Design may explore strategies like subtle nudges, self-awareness promotion, and personalized alerts to support their self-reliant nature [23].	Collectivists place a strong emphasis on interdependence and collective interests [97], valuing relationships and social integration [10, 71]. Design can explore help-oriented approaches that leverage social support networks, potentially promoting collaborative group activities to reduce feelings of alienation and helplessness.
	Individualistic individuals readily share personal information with practitioners, emphasizing professional support over group affiliation [2]. Design interventions should focus on enabling and supporting direct and transparent communication between individuals and healthcare practitioners.	Collectivists prioritize family interests, resulting in low rates of seeking help from external sources due to family intrusiveness and mental health stigma. From a design perspective, the focus should be on privacy, confidentiality, and trust to address these concerns. Additionally, exploring ways to foster support and awareness within families and communities is essential [74].
	Individualists, preferring direct and formal conversations, usually share only essential information with doctors, avoiding unnecessary personal details. Design interventions could thus focus on straightforward, efficient communication tools that respect their preference for brevity and formality in medical interactions [77].	Collectivists value doctors who take a familial, personal interest in their lives, aligning with their collectivistic values. Without this personal touch, they may see the doctor as detached. Design solutions could therefore emphasize personalization and warmth in patient-doctor interactions, fostering a sense of care and community [77].
Info Exchange	For individualists, who often have smaller support networks and may hesitate to seek help for personal issues, technology can offer alternative support. Platforms like conversational agents can be effective, catering to their preference for informative content on social media. Design strategies should focus on understanding online help-seeking behaviors, developing platforms that are informative yet static [4, 51, 55, 80, 93].	Collectivists, facing challenges like social stigma in adhering to mental health interventions, may benefit from engaging designs incorporating gamification and reward structures. Given their more interactive and emotionally driven use of social media, design interventions could utilize these platforms for data collection and early detection of mental health issues, enhancing motivation and compliance [4, 23, 23, 51, 55].
UI Design Implication		
Digital support	Content should adopt a 'you-attitude,' focusing on images of people achieving goals, such as positive treatment outcomes, to inspire and motivate [39].	Content in a more indirect and informal style is preferred, but it should still offer comprehensive support for any claims made [39].
	Prefer rational appeal [16], thus designs should focus on clear, data-driven content. They should offer scientifically robust support for claims and explanations for treatment options, emphasizing logic over emotion	Favor social and emotional appeal would benefit from imagery and content that foster empathy, understanding, and connection [16]. Incorporating patient reviews and experiences can provide persuasive support for claims, highlighting the co-creation of experiences and shared perceptions.
	Interface should provide a wide range of options should be provided, such as dropdown menus and checkboxes, without presuming default selections [39].	Interface should prominently feature links to various organizations, institutions, or affiliated groups, showcasing the strength and breadth of these networks [39].
	Cultures valuing efficiency prefer clutter-free graphics and well-designed interfaces[39, 61]. They often favor interfaces with monochromatic color schemes [6]	Tend to prefer interfaces with traditional design and human-centric imagery [61]. They are also inclined towards colorful and bright interface designs [6].
	Prefer a high text-to-image ratio [39]	Prefer a high image-to-text ratio [39].
	Prefer low multimodality [44]	Prefer high multimodality [44]

Table 1: Design implications of the cultural dimension IvC

Power Distance		
	High Power Distance	Low Power Distance
Behavioural Design Implication		
Power Inequality	Respect and value the authority of medical staff [45, 48, 91]. Design interventions could explore emphasizing the expertise and knowledge of medical professionals, creating a platform that enhances collaboration between medical staff and patients while primarily featuring content developed by healthcare experts.	Not strongly inclined towards medical staff's authority [45, 48, 91]. Design should strike a balance between professional medical content and user-generated contributions. Platforms that facilitate both expert advice and community collaboration allow for a more egalitarian exchange of information and opinions.
Expectation	Anticipate therapists to adopt a directive approach and provide advice. Consequently, when therapists fail to meet this expectation, they run the risk of being perceived as incompetent or indifferent [41, 46]. Tools can guide therapists to provide clear advice and maintain an authoritative stance, while acknowledging the clients' preference for clear direction without heavy involvement in decision-making [29].	Consider it ideal to involve all those who will be directly affected by a decision in the decision-making process [49]. Design could facilitate a more collaborative approach. Interventions can provide platforms for group consultation and joint decision-making [2].
Self-Disclosure	The idea of openly discussing private problems with strangers may be seen as challenging or even inappropriate. Instead, there may be a preference for maintaining the privacy and seeking guidance and support within the family or close social circles [2]. From a design perspective, design could focus on privacy and family or close social circle involvement. Tools and platforms can facilitate confidential communication and support within trusted networks, respecting cultural norms of privacy and internal support systems.	Tend to experience less discomfort discussing private matters with strangers, due to a focus on equality and a casual attitude towards authority [2]. Designs could further encourage open communication with professionals. Platforms that facilitate straightforward, equal-footing interactions can capitalize on the casual attitude towards authority, promoting transparency and ease in discussing private matters.
Collaboration	Tend to perceive doctors as the utmost authority figure and follow their words with complete obedience [77, 91], delegating decision-making to them regarding their health. Design interventions could explore strategies to bridge the gap between patient and doctor, while honoring their preference of light-involvement in the decision-making process. Tools might also offer alternative methods for symptom elicitation and support self-agency.	Reduced PD positively impacts communication and participation in consultations. The doctor takes more of a guiding and advisory role whereas the decision-making is conducted by the patient or together. They do not expect their doctor to know it all and disagreements are seen as an opportunity [77]. Design can act as a collaborative medium to allow the doctor and patient to explore and uncover underlying issues together.
UI Design Implication		
	Constrain interactions to linear navigation, few links, minimize navigation possibilities [13, 23, 39, 54, 61] Little information should be provided at the first/highest level. Information is structured hierarchally [13, 61]. Interfaces should employ strict error messages with a demanding tone [61] Provide various opportunities for support (chat-based, email contact, community forums) [61]. Images should display authority figures (doctors, nurses, therapists) [39, 61]	Nonlinear navigation and different access and navigation possibilities can be implemented, and should be used to enable exploration [13, 23, 39, 54, 61] Most information should be provided at the interface level. The hierarchy of information is flat [13, 61]. Interfaces should use friendly error messages [61] Due to the rare need for support, focus on support is secondary and only one or a few support possibilities can be provided [61]. Images should display individuals from everyday life [39, 61].

Table 2: Design implications of the cultural dimension PD

Time Orientation		
	Long-term Orientation	Short-term Orientation
Behavioural Design Implication		
Attribution	Inclined to attribute success to effort and failure to a lack of it. From a design perspective, interventions should focus on fostering persistence and sustainable improvements [47]. These designs could aid individuals with a long-term orientation by helping them visualize their persistence and manage feelings like shame and guilt during treatment.	Inclined to attribute both failure and success to luck or fate [48]. May show less motivation for sustained care measures, as they might not see immediate benefits of preventive actions. Design interventions in this context could emphasize immediate outcomes [47] and the impact of personal effort. By visualizing continuous improvements and progress, these designs can also highlight and visualize the long-term benefits [67].
Preventative Care	Individuals valuing follow-up care and engaging in self-screening can benefit from design interventions that adopt an advisory and supportive role [37, 76]. Such designs could focus on empowering individuals with information and tools necessary for making informed health decisions, emphasizing the importance of continuous self-care and awareness.	Less concerned with follow-up care and self-screening in the absence of immediate symptoms [76]. Design interventions need to assume a more persuasive role. These should be geared towards motivating individuals to engage in early disease detection. Effective designs could highlight the immediate and long-term benefits of preventive measures, demonstrating the value of consistent self-screening and follow-up care in maintaining overall health..
Habits	More inclined to partake in health-promoting activities such as regular exercise and maintaining a balanced diet [40, 58]. From a design perspective, simple and accessible digital resources could effectively reinforce and encourage their already positive health habits.	More inclined to partake in pleasurable health risks such as smoking and consuming alcohol or unhealthy eating behaviour [40, 76]. These interventions should focus on motivating health-promoting behaviors and improving adherence, utilizing interactive and engaging digital platforms to initiate and sustain positive lifestyle changes[42].
Traditions	Individuals who view traditions as adaptable to changing circumstances may be more receptive to innovative design approaches[48]. From a design perspective, this openness suggests a greater acceptance of experimental and holistic methods. Design interventions could, therefore, emphasize flexibility and innovation, incorporating cutting-edge techniques and holistic practices to align with this cultural adaptability.	Tend to have immense respect for traditions and often adopt a passive role, expecting clear guidance in what to do[15]. Design interventions should aim to harmonize traditional practices with contemporary healthcare. This could involve validating and integrating accurate traditional beliefs into modern treatment plans. Additionally, interventions might leverage structured rules and guidelines as credible sources of information, aligning with cultural preferences for clear direction and authority [47].
Social Network	Are more likely to invest in social networks or interpersonal relations with acquaintances[65]. Believe that their personal network provides resources for achievement or success [59]. Consequently, design interventions could focus on enhancing interpersonal relations and integrating the patient's close social network into their treatment.	Their focus on "saving face" may influence how an individual approach rehabilitation or treatment. This cultural aspect can lead to feelings of shame or a perceived need to conceal disabilities, diseases, or illnesses [48]. In response, design should prioritize discreet, personalized approaches that ensure anonymity and privacy.
Causality	More disease-oriented and are more likely to ask "What" and "How" than "Why" [48].From a design perspective, the focus should be on developing tools that facilitate efficient diagnosis and treatment, providing clear and actionable information.	More symptom-oriented and are mostly interested in the "Why" [48].From a design perspective, the focus should be on creating platforms that encourage in-depth exploration of symptoms' underlying causes and foster patient education and self-awareness.
UI Design Implication		
	Interface navigation style and content can be more complex as users persevere until they understand. Interventions should contain content that is of practical value and can use relationships to verify the credibility of the information [45, 54]. Content highly structured into small units [60, 61] Interface design should be designed in a timeless and classic way, without frequent changes to align to design trends [59]. An emphasis on using imagery which highlights people [59].	Interface design should be structured in a way that allows users to complete tasks quickly. Prefer a reduced information density [54, 85]. Rules should be used to verify the credibility of information and information content should be based on truth and certainty of beliefs [47]. Content can be arranged around a focal area [60, 61] Interface design that is appropriate and up-to-date for now, and likely to be outdated when new design trends begin [59, 61]. Focus on only showing the relevant tasks or products [59].

Table 3: Design implications of the cultural dimension LTO

Uncertainty Avoidance		
	Low Uncertainty Avoidance	High Uncertainty Avoidance
Behavioural Design Implication		
Trust & Innovation	Tend to exhibit greater openness to innovative treatment approaches [8]. They may also display a willingness to explore alternative and complementary therapies. From a design perspective, focusing on innovation becomes pivotal, as these individuals require fewer rules, are quick to embrace and trust technology, and exhibit curiosity towards innovative solutions [47, 59].	Typically have a lower tolerance for ambiguity and prefer established protocols and traditional treatment methods [63]. Their trust in technology tends to be low [56]. From a design perspective, focus could be on building trust, acceptance and ensuring security by emphasizing effectiveness and success through use cases, user reviews, and demonstrations. Striking a balance between tradition and innovation is essential, with a primary goal of achieving adoption and acceptance [8, 30, 59].
Proactiveness	Due to their generally low stress and anxiety levels, individuals tend to oversee symptoms, avoid antibiotics and prescription drugs, and adopt a 'wait-and-see' approach to health issues [11, 89]. From a design perspective, the focus could be on supporting users in their decision-making process, help users balance the decision between waiting and taking action regarding symptoms, and facilitating or automating the recognition of symptoms.	Tend to exhibit high anxiety and stress levels [63], prompting them to seek extensive information, second opinions, and explanations to alleviate uncertainty and minimize potential risks [24]. Also tend to use prescription drugs inappropriately due to their perceived risk of doing nothing [25]. From a design perspective, the focus could be on minimize uncertainty through educating and informing users adequately about their health concerns and options, while facilitating effective knowledge transfer, increased control over health records, and access to resources [59].
Communication	Tend to prioritize eye contact and rapport-building with their doctors, resulting in more open doctor-patient relationships and psychosocial information exchange [89]. From a design perspective, it could mean for technology to take on a secondary role during the consultation, allowing the doctor-patient relationship to remain the primary focus. Technology should could and enhance communication without becoming a central distraction, preserving the importance of face-to-face interaction.	Tend to feel uncomfortable when directly asked questions about their health, leading to difficulties in expressing their concerns [89]. They also pay less attention to building rapport in consultations and are less likely to bring up concerns or detect cues. From a design perspective, technology can serve as a facilitator and shared medium for communication between individuals and their healthcare providers, offering alternative options [84] to traditional face-to-face interactions. Focus could be on making interactions, social cues, and information exchange more explicit.
Empowerment	Tend to attribute results and treatment outcomes to their own abilities or shortcomings [47]. From a design perspective, it is crucial to emphasize the multifaceted nature of health outcomes and educate patients about various contributing factors beyond individual control. Interaction should be thoughtfully crafted to minimize negative framing or wording that may evoke feelings of personal failure or guilt and support individuals in coping with any feelings of personal shortcomings or guilt.	Tend to feel more powerless when it comes to their health and attribute outcomes to external forces [47]. From a design perspective, interventions could focus on highlighting users about the benefits of treatment, preventive and follow-up care while also emphasizing the influence of individuals' efforts in fostering agency in managing their health. The design could focus on empowering individuals to take control of their health outcomes.
UI Design Implication		
	Interface design should prioritize complexity with maximal content and choices to allow for exploration [59, 61]. Most information should be displayed at the interface level. Interface design should prioritize coding of color, shape, and texture cues used to maximize information rather than redundancy [59–61]. Design should prioritize the content, with the emphasis on providing users with comprehensive information, with task orientation being secondary [59, 60]	Interface design should focus on simplicity, with clear metaphors, limited choices, and restricted amounts of data [45, 61]. Information should be organized hierarchically. Interface design should focus on using redundant cues such as sound, color, typography, and more to carefully encode meaning [59–61]. Design should prioritize task orientation. This entails providing clear guidance, error messages, continuous feedback, and support for users as they engage in specific tasks. [59, 61].

Table 4: Design implications of the cultural dimension UA

Masculinity vs. Femininity		
	Masculinity	Femininity
Behavioural Design Implication		
Attitudes	Tend to be reluctant to seek medical care or partake in preventative care, perceiving unhealthy behaviors as signs of strength, self-reliance, and independence [18, 31]. They also lean towards short-term therapy and group-based workshops [75]. From a design perspective, the focus could be on addressing these tendencies by supporting healthy behavior and behavior change. From a design perspective, prioritizing communication that resonates with their values of strength and self-reliance while gently encouraging a more open and proactive attitude towards healthcare, should be a key consideration.	Individuals tend to exhibit a more positive help-seeking attitude and are more inclined to seek out medical assistance, participate in preventive care and utilize holistic approaches [3, 31, 31]. From a design perspective, technology can provide alternatives to traditional treatments, which can either support or extend traditional approaches, with a primary goal of offering additional support for those who wish to explore complementary healthcare options.
Approach	Tend to prefer action-focused interventions based on problem-solving strategies, emphasizing the practical methods over talk-based therapy[28]. The use of decision trees and exploratory questions to promote action, as well as case studies and extensive psychoeducation reframing depression as a medical condition, were promoted as useful treatment components [12]. From a design perspective, the focus could be on developing interventions that align with their preference and rationality for action-oriented strategies and emphasize the practical utility of therapy, incorporating decision-support tools, and problem-solving modules.	Value professionals who listen and make them feel 'heard,' as they often perform more 'emotional work' in their daily relationships, such as providing emotional support[28]. From a design perspective, the focus could be on creating and supporting environments, physical or digital, that values active listening, empathy, and emotional support. Designing interventions that prioritize building trusting relationships, where individuals feel truly heard and understood, should be a key consideration.
Engagement	Attention and adherence is gained through games and competitions [45, 59]. From a design perspective, interventions could explore the use of gamification or games for treatment and increased adherence.	Mutual exchange and support are more important than functionality or mastery [59]. From a design perspective, interventions could prioritize functionalities that facilitate exchange, communication, and patient connections with others in similar situations.
UI Design Implication		
	Focus on encouraging words in its communication with users [14, 26, 45]. Interaction should be individual-oriented [34]. Tend to prefer highly contrasting and bright colours [26, 90]. Interfaces should minimize navigation possibilities and choices [34].	Focus on the emotional and personal presentation of content [14, 26, 45]. Interaction should be group-oriented [34]. Tend to prefer little saturation and pastel colours [26, 90]. Interfaces should provide ample exploration and navigation possibilities, offering users multiple choices [34].

Table 5: Design implications of the cultural dimension MvF